

SYDNEY A. JENKINS

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EDUCATION

Massachusetts Institute of Technology (MIT)

2021 - present

Ph.D. Candidate in Physics

Advisor: Dr. Andrew Vanderburg

Research interests: White dwarf planetary systems, white dwarf pollution, exoplanet atmospheres, and exoplanet compositions

University of Chicago

2017 - 2021

Bachelor of Arts in Physics and Bachelor of Science in Computer Science

PUBLICATIONS

First-Author Publications

4. **S. Jenkins**, et al. *Atmospheric Signatures of Common Envelope Evolution in White Dwarf Planets*. Submitted to ApJ.
3. **S. Jenkins**, et al. 2025. *An Eccentric Sub-Neptune Moving Into the Evaporation Desert*. ApJL, 989, L20. DOI: 10.3847/2041-8213/adea46
2. **S. Jenkins**, et al. 2024. *Absence of a Correlation between White Dwarf Planetary Accretion and Primordial Stellar Metallicity*. MNRAS, 532, 394. DOI: 10.1093/mnras/stae1506
1. **S. Jenkins**, et al. 2021. *VLT Spectroscopy of Ultra-Faint Dwarf Galaxies: Boötes I, Leo IV, Leo V*. ApJ, 920, 92. DOI: 10.3847/1538-4357/ac1353

Co-Author Publications

16. M. A. Kenworthy et al. (10 co-authors including **S. Jenkins**) 2026. *Upper limits on exosatellites around Beta Pictoris b*. Accepted to MNRAS. DOI:10.48550/arXiv.2606.04685
15. Giacomo Mantovan et al. (20 co-authors including **S. Jenkins**) 2026. *A tidally detached super Neptune on a strongly misaligned retrograde orbit*. A&A, 709, L15. DOI:10.1051/0004-6361/202660217
14. Y. Wu et al. (21 co-authors including **S. Jenkins**) 2026. *Detection of four cold Jupiters through combined analyses of radial velocity and astrometry data*. AJ, 171, 189. DOI:10.3847/1538-3881/ae40f5
13. B. Macias, **S. Jenkins**, et al. 2026. *First Astrometric Limits on Binary Planets and Exomoons orbiting β Pictoris b*. AJ, 171, 197. DOI:10.48550/arXiv.2512.00160
12. Madyson Barber et al. (17 co-authors including **S. Jenkins**) 2026. *TESS Investigation - Demographics of Young Exoplanets (TI-DYE) IV: a giant planet orbiting a 34 Myr Sun-like star in the Vela association*. AJ, 171, 20. DOI:10.3847/1538-3881/ae1b91
11. N. Aldarondo Quiñones, **S. Jenkins**, et al. 2025. *Reassessing the Relationship Between Stellar X-ray Luminosity and Age with eROSITA Data Release 1*. PASP, 137, 114206. DOI:10.1088/1538-3873/ae1e2c

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10. Robert Aloisi et al. (46 co-authors including **S. Jenkins**) 2025. *Searching for Exoplanets Born Outside the Milky Way: VOYAGERS Survey Design*. PASP, 137, 114404.
DOI:10.1088/1538-3873/ae1cf0.
9. P. Noti, E. Lee, D. Kitzmann, R. MacDonald, **S. Jenkins**, et al. 2025. *Modelling the 3D atmospheric structure of the cold Jupiter WD1856+534b orbiting a white dwarf*. MNRAS, staf1119. DOI:10.1093/mnras/staf1119
8. M. A. Limbach, A. Vanderburg, R. MacDonald, K. Stevenson, **S. Jenkins**, et al. 2025. *Thermal Emission and Confirmation of the Frigid White Dwarf Exoplanet WD 1856+534 b*. ApJL, 984, L28. DOI:10.3847/2041-8213/ad74ed
7. M. A. Limbach, A. Vanderburg, A. Venner, S. Blouin, K. Stevenson, R. MacDonald, **S. Jenkins**, et al. 2024. *The MIRI Exoplanets Orbiting White Dwarfs (MEOW) Survey: Mid-Infrared Excess Reveals a Giant Planet Candidate around a Nearby White Dwarf*. ApJL, 973, L11.
DOI:10.3847/2041-8213/ad74ed
6. M. Coughlin, et al. (52 co-authors including **S. Jenkins**) 2023. *A data science platform to enable time-domain astronomy*. ApJS, 267, 31. DOI:10.3847/1538-4365/acdee1
5. D. Tanoglidis, A. Ćiprijanović, A. Drlica-Wagner, B. Nord, M. Wang, A. J. Amsellem, K. Downey, **S. Jenkins**, et al. 2022. *DeepGhostBusters: Using Mask R-CNN to Detect and Mask Ghosting and Scattered-Light Artifacts from Optical Survey Images*. Astronomy and Computing, 39, 100580. DOI: 10.1016/j.ascom.2022.100580
4. A. Ćiprijanović, D. Kafkes, K. Downey, **S. Jenkins**, et al. 2021. *DeepMerge II: Building Robust Deep Learning Algorithms for Merging Galaxy Identification Across Domains*. MNRAS, 506, 677. DOI: 10.1093/mnras/stab1677
3. A. Ćiprijanović, D. Kafkes, **S. Jenkins**, et al. 2020. *Domain Adaptation Techniques for Improved Cross-Domain Study of Galaxy Mergers*. 34th Conference on Neural Information Processing Systems. DOI: 10.48550/arXiv.2011.03591
2. T. Kurihana, I. Foster, M. Maire, R. Willet, **S. Jenkins**, et al. 2019. *Cloud Characterization with Deep Learning II*. 9th International Workshop on Climate Informatics. arXiv:2209.15585
1. T. Li, J. Simon, K. Kuehn, A. Pace, D. Erkal, K. Bechtol, B. Yanny, A. Drlica-Wagner, J. Marshall, C. Lidman, E. Balbinot, D. Carollo, **S. Jenkins** et al. 2018. *The First Tidally Disrupted Ultra-Faint Dwarf Galaxy? - Spectroscopic Analysis of the Tucana III Stream*. ApJ, 866, 22.
DOI: 10.3847/1538-4357/aadf91

ACCEPTED OBSERVING PROPOSALS & OBSERVING EXPERIENCE

Accepted Observing Proposals

JWST	2025
<i>Probing the Dynamical History of a White Dwarf Planet</i>	
Magellan	2023, 2024, 2025
<i>Characterizing Extragalactic Worlds using Polluted White Dwarfs</i>	
Magellan	2024
<i>TOI-5800 b: A sub-Neptune planet candidate ideal for JWST follow-up</i>	

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Ground-Based Observing Experience

TNG/HARPS-N , 8 nights	2025
Magellan Clay/MIKE , 1 night	2025
Magellan Clay/PFS , 6 nights	2025
Magellan Baade/MagE , 1 night	2024
TNG/HARPS-N , 10 nights	2024
Magellan Baade/MagE , 2 nights	2023

PRESENTATIONS

- Talk** *WD 1856 and the Search for Frigid Worlds with JWST*. MIT Kavli Institute Journal Club, Cambridge, MA, 4/2026
- Talk** *Probing the Dynamical History of White Dwarf Planets: Atmospheric Signatures of Common Envelope Evolution*. Boston Area Planetary Science Meeting, Boston, MA, 1/2026
- Talk** *Probing the Dynamical History of White Dwarf Planets: Atmospheric Signatures of Common Envelope Evolution*. American Astronomical Society Meeting, Phoenix, AZ, 1/2026
- Talk** *Probing the Population of Giant Planets around White Dwarfs with JWST*. 51 Pegasi b: Cool Giant Planets and their Systems, Saint-Michel-l'Observatoire, France, 10/2025
- Talk** *Planetary Evolution Across Stellar Lifetimes*. Department talk, University of Cambridge, Cambridge, UK, 9/2025
- Talk** *Planetary Evolution Across Stellar Lifetimes*. Department talk, University of Warwick, Coventry, UK, 9/2025
- Talk** *An Eccentric Sub-Neptune Moving Into the Evaporation Desert*. The Fate of Neptunian Worlds: Exploring Demographics, Atmospheres and Evolution, Coventry, UK, 9/2025
- Talk** *Discovery and Characterization of Planets Around White Dwarf Stars*. OWL Summer Program, Santa Cruz, CA, 7/2025
- Talk** *Eccentric Sub-Neptune Approaching the Photoevaporation Desert*. 2025 Magellan Science Meeting, Washington, D.C., 5/2025
- Talk** *Eccentric Sub-Neptune Approaching the Photoevaporation Desert*. CfA Exoplanet Pizza Lunch, Boston, MA, 3/2025
- Talk** *Thermal Emission from the First Giant Planet Transiting a White Dwarf*. TESS Science Conference III, Boston, MA, 8/2024
- Talk** *Thermal Emission from the First Giant Planet Transiting a White Dwarf*. Emerging Researchers in Exoplanet Science IX, Ithaca, NY, 7/2024
- Poster** *Thermal Emission from the First Giant Planet Transiting a White Dwarf*. Exoplanets 5, Leiden, The Netherlands, 6/2024
- Talk** *Thermal Emission from the First Giant Planet Transiting a White Dwarf*. Extreme Solar Systems, Christchurch, New Zealand, 3/2024
- Poster** *Thermal Emission from the First Giant Planet Transiting a White Dwarf*. Open Problems in the Astrophysics of Gas Giants, Puerto Natales, Chile, 12/2023

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- Poster** *Constraining White Dwarf Pollution Mechanisms with the GMT*. Ninth Annual Giant Magellan Telescope Community Science Meeting, Washington, D.C., 9/2023
- Poster** *Constraining White Dwarf Pollution Mechanisms with the HWO*. Science with the Habitable Worlds Observatory and Beyond, Baltimore, MD, 7/2023
- Poster** *Probing the Role of Jupiter Analogs in White Dwarf Pollution*. ERES, New Haven, CT, June 6/2023
- Talk** *Direct Imaging Exoplanets with JWST*. MIT Kavli Institute Journal Club, Cambridge, MA, 11/2022
- Talk** *Probing the Role of Jupiter Analogs in White Dwarf Pollution*. CfA Exoplanet Pizza Lunch, Boston, MA, 10/2022
- Poster** *Probing the Role of Jupiter Analogs in White Dwarf Pollution*. Sagan Workshop, Los Angeles, CA, 8/2022
- Poster** *Taxonomy-Aware Neural Network Classification of Variable Stars*. American Astronomical Society Meeting, 1/2021
- Talk** *Identifying High-Redshift Galaxy Mergers with Domain Adaptation*. New Perspectives, Batavia, IL, 7/2020
- Poster** *Probing Dark Matter Halos with Ultra-Faint Dwarf Galaxies: A Spectroscopic Analysis of Leo V*. American Physical Society March Meeting, 3/2020
- Poster** *The Kinematics of Ultra-Faint Dwarf Galaxies: A Case Study of Spectroscopic Analyses*. American Astronomical Society Meeting, Honolulu, HI, 1/2020
- Poster** *Developing Unsupervised Learning Models for Cloud Classification*. American Geophysical Union Fall Meeting, San Francisco, CA, 12/2019

HONORS & AWARDS

NSF GRFP	2021
Most Accessible Talk Award (<i>New Perspectives Conference</i>)	2020
Metcalf Grant Recipient (<i>University of Chicago</i>)	2020
Dean's List (<i>University of Chicago</i>)	2017-2019
Provost's Scholar (<i>University of Chicago</i>)	2017

RESEARCH SUPERVISION

Nadja Aldarondo Quiñones	2024 - 2025
<i>MIT MSRP Student (co-supervised with Andrew Vanderburg and Melinda Soares-Furtado)</i>	
Isabella Macias	2023 - 2025
<i>MIT MSRP Student (co-supervised with Andrew Vanderburg)</i>	

TEACHING

TA for Physics I	2024
<i>Department of Physics, MIT</i>	
TA for Inclusive Technology: Design for Underserved and Marginalized Communities	2021
<i>Department of Computer Science, University of Chicago</i>	

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MENTORSHIP & OUTREACH

Graduates Advising Graduate Admissions Physics Department	2024 - present
TESS Volunteer NASA STEM Day at Fenway	2024, 2025
Mentor MIT Graduate & Undergraduate Womxn in Physics	2023 - present
Project Leader Warrior-Scholar Project	2023, 2025
<i>Led weeklong course on exoplanet science for team of veterans</i>	
Speaker Birches Middle School	2023
<i>Speaker at outreach event for middle school students</i>	
MIT Sidewalk Astrogazers Member MIT Kavli Institute	2021 - present
Mentor MIT Physics Graduate Application Assistance Program	2021 - present
Mentor MIT Physics Mentor Program	2021 - present
Organizing Committee Member Expand Your Horizons	2019 - 2021
<i>Organized annual symposium to engage middle school girls in STEM</i>	